

Sr.No.4107

**Master in Computer Application (MCA) (Two Years Programme) -
1st Sem. (Credit Based Evaluation & Grading System) (2024-26)
(2224)**

Paper: CSL-4053 System Simulation

Time Allowed: 3hrs.

Max. Marks: 70

Note: Attempt five questions in all by selecting at least one question from each section. Each question carries equal marks.

SECTION A

1. a) What are the different types of models? What are the principles used in modeling? (7)
b) Differentiate between continuous and discrete systems by giving examples. (7)
2. a) Distinguish between: Fixed time step and event to event model. (7)
b) Why are random numbers needed in simulation? Explain rejection method for generation of random numbers. (7)

SECTION B

3. a) Derive poisson distribution formula. (10)
b) In a local bank, with 1 teller, average arrival rate is 5 customers per hour and average service rate is 6 customers per hour.
Find : (i)utilization factor (ii) average number of customers in the system. (4)
4. Explain two server queue with help of a flowchart (14)

SECTION C

5. Discuss inventory control simulation example in which demand is Poisson distributed and procurement lead time is Erlang distributed. (14)
6. a) What are the various methods used in forecasting? (7)
b) Discuss validation of models used in simulation. (7)

SECTION D

7. a) What are the important characteristics of expression based languages? (7)
b) What are the factors in selection of discrete system simulation language? (7)
8. Write short notes on:
(a) GPSS (7)
(b) SIMSCRIPT (7)

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